Semantic-Domain Differential Compression for Real-Time V2X Communication

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Abstract. Semantic scene sharing for collaborative perception in autonomous of semantic information via bandwidth-limited V2X links. Existing techniques either send high-resolution segmentation masks or apply on learned compression techniques leading to decoding costs at the receiver. In this paper we introduce CARLE (Class-Aware Run-Length Encoding), a simple, deterministic compression scheme of the semantic domain that only represents label changes between frames without estimation, or models. Consecutive segmentation masks are compared on a pixel level; areas where the class label has changed are filled with the label of the current frame, while positions a reserved null symbol (0), to create a sparse change map, which is as start-length-class triplets. The receiver maintains a semantic world map that is updated by the change-patches, in line with goal-oriented collaborative perception. Evaluated across three datasets under CPU-only constraints2,424 nuScenes Mini frames, 1,073 field-test frames, and 1,487 Cityscapes panoptic pairsCARLE achieves 265.1× at 60.40 ms, 884× at 17.97 ms, and 179,346× at 179.44 ms respectively, with mIoU = 1.00 and no decoder requirement.

Keywords: Semantic Communication· V2X Networks· Autonomous Vehicles· Semantic Segmentation· Differential Compression

1 Introduction

Autonomous vehicles (AVs) navigating dense and unstructured trafficcharacteristic of Asia-Pacific environmentsrely on real-time semantic scene understanding for safe manoeuvring. Collaborative driving, where vehicles share perceptual representations to improve situational awareness, has emerged as a key paradigm for handling scenarios beyond single-sensor reach [1, 2]. The standard approach transmits full-resolution semantic segmentation masks over Vehicle-to-Everything (V2X) links. Under 3GPP TR 38.885 in urban microcell constraints, this is prohibitively bandwidth-intensive [3]: a single 900×1600 mask at 8 bits/pixel requires 11,520 Kb per frame, and modern AVs carry 6 or more cameras, yielding a combined raw bitrate exceeding 69,120 Kb/frame.

Existing approaches address this in two ways, both with significant limitations in all of them. Communication-efficient collaborative perception schemes such as Where2comm [4] and feature compression methods based on heavy backbones such as ResNet-101 or PointNet++ [1] reduce raw data volume but require full mask reconstruction or deep decoding at the receiver, imposing latency



Fig. 1: CARLE on-vehicle semantic compression pipeline.

and compute overhead impractical on embedded edge nodes. Generic lossless codecs LZ4r [5] and SnappyR [6] achieve high throughput via byte-level match-finding inherited from the Lempel–Ziv family [7], while video codecs DVC [8] and VVC [9] exploit temporal dependencies using motion estimation and residual coding in pixel space. Critically, none of these exploit the discrete structure of semantic label maps, and all treat each frame independently or operate on raw pixel valuesignoring the dominant source of redundancy in driving scenes. Goal-oriented semantic communication [10–12] advocates transmitting only task-relevant information, yet existing instantiations still rely on learned models or frame-independent codecs that discard temporal structure.

A key observation motivating this work is *temporal sparsity*: the vast majority of pixels carry the same semantic label across consecutive frames. Our empirical analysis across three datasets confirms this at scale. In 10 nuScenes Mini scenes, only 4.5% of pixels change class on average (median 1.0%). Among 1,073 real-world field-test frames, 48.8% exhibit zero semantic change and the median active changed-pixel fraction (δ_{pix}) is just 0.35%. Across 1,487 Cityscapes panoptic mask pairs at 1024×2048, only 0.007% of pixels differ on averageconfirming that structured semantic representations are overwhelmingly temporally stable even at high resolution. Lightweight segmentation architectures tailored for embedded deployment [13, 14] produce exactly the discrete label maps whose temporal stability CARLE is designed to exploit.

We propose CARLE (Class-Aware Run-Length Encoding), a deterministic, training-free compression framework that encodes only the pixels whose semantic class has changed between frames, preserving the *current* class label at changed locations and zeroing unchanged regions. The transmitted payload is precisely the *semantic world-model update* needed by the receivera direct instantiation of goal-oriented cooperative perception [11]. The contributions of this paper are:

- A class-preserving null-label differential operator that maps pixel-wise label changes to a sparse representation: changed positions carry the current frame's label; unchanged positions are assigned a reserved null symbol (0), unambiguously separating "no change" from "changed to class c" without motion estimation or learned models.
- CARLE encoding of the sparse change map as start–length–class triplets, achieving 265×–179,346× compression at 17.97–179.44 ms (CPU-only), meeting the 3GPP V2X 100 ms constraint for field-test and nuScenes resolutions without GPU acceleration.
- Empirical confirmation of scene-adaptive bandwidth across nuScenes Mini, real-world road videos, and Cityscapes panoptic masks (18 cities): CR scales inversely with changed-pixel fraction across four orders of magnitude without dataset-specific tuning, consistent with Eq. (5).

2 Methodology

2.1 Scene Simplification

Let the input video be a sequence of frames $\{I_t\}_{t=1}^{T}$, where $I_t \in \mathbb{R}^{H\times W\times 3}$. Each frame is processed by a pre-trained segmentation model $\mathcal{S}$ to produce a full label map $F_t \in \mathbb{Z}^{H\times W}$. To remove semantic noise and focus transmission on safety-critical content, all classes are projected onto:

$$\mathcal{C}_{\text{critical}} = \{c_1 = \text{road},\ c_2 = \text{vehicle},\ c_3 = \text{pedestrian},\ c_4 = \text{other}\} \tag{1}$$

yielding a simplified mask $\tilde{F}_t$ where non-critical pixels map to c_4. This four-class representation retains all collision-relevant semantics while reducing label entropy.

2.2 Class-Preserving Differential Map

Given two consecutive simplified masks $F_t, F_{t+1} \in \mathbb{Z}^{H\times W}$, where class labels $\in \mathcal{C}_{\text{critical}} = \{1, 2, 3, 4\}$ and 0 is reserved as the null symbol, we define the *semantic difference map*:

$$D_t(i,j) = \begin{cases} F_{t+1}(i,j), & \text{if } F_{t+1}(i,j) \neq F_t(i,j) \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

When a pixel's label changes between frames, the result carries the *updated* class label from the current frame F_{t+1}not a bitwise residual. When a pixel's label is unchanged, the result is set to zero (0). This is illustrated in Fig. 2: if a vehicle (class 2) moves right, the pixel it vacated now belongs to road (class 1), so that position writes $D_t = 1$; the new vehicle position writes $D_t = 2$; pixels that remain road throughout write $D_t = 0$ (null). The receiver can therefore directly apply every non-zero entry to its persistent semantic map without any additional decoding step.

At the receiver, the persistent semantic map is initialised from the first full frame and updated by writing each triplet (s_k, l_k, c_k) into the corresponding pixel positions; all positions absent from $\mathcal{R}_{t+1}$ retain their previous label.

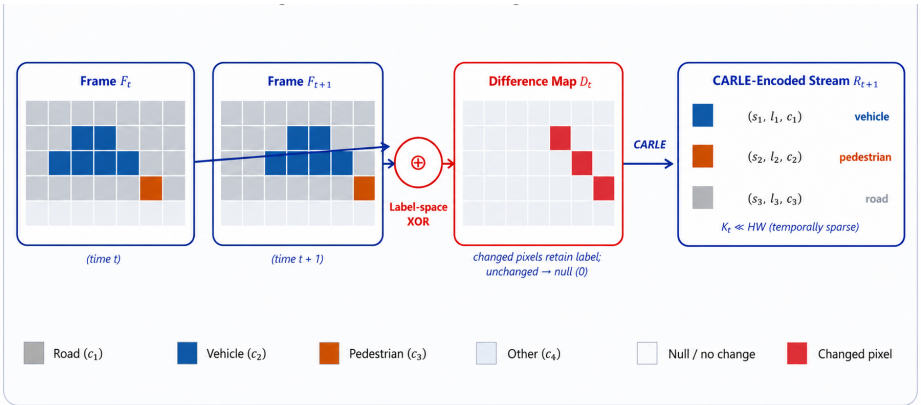


Fig. 2: Class-preserving differential encoding.

2.3 Class-Aware Run-Length Encoding (CARLE)

The 2D difference map D_t is flattened row-wise into a 1D vector $\mathbf{d}_t = (D_t) \in \mathbb{Z}^{HW}$. Contiguous non-zero segments sharing the same class label are grouped into triplets:

$$\mathcal{R}_k = (s_k,\ l_k,\ c_k) \tag{3}$$

where s_k is the start index, l_k the run length, and $c_k \in \mathcal{C}_{\text{critical}}$ the class label. The full compressed frame is:

$$\mathcal{R}_{t+1} = \{\mathcal{R}_k\}_{k=1}^{K_t}, \quad K_t \ll HW \text{ for temporally stable scenes} \tag{4}$$

2.4 Compression Efficiency Analysis

Let the uncompressed mask size be $B_{\text{raw}} = HW \cdot \log_2 C$ bits. The compressed size is:

$$B_{\text{CARLE}} = K_t \cdot (\log_2 HW + \log_2 L_{\max} + \log_2 C) \tag{5}$$

where $L_{\max}$ is the maximum run length. The compression ratio is:

$$\text{CR} = \frac{B_{\text{raw}}}{B_{\text{CARLE}}} \gg 1 \quad \text{when } K_t \ll HW \tag{6}$$

Bandwidth scales directly with scene dynamics: in stable scenes $K_t \to 0$ and compression is maximal; in motion-rich scenes K_t grows but remains bounded by the number of class boundary transitions, keeping bandwidth tractable without any explicit rate-control mechanism. In the degenerate case where all pixels change ($K_t = HW$), B_{CARLE} approaches B_{raw} and CR $\to 1$; however, this scenario does not arise in practice for structured semantic label maps, as confirmed by the 0.0074.5% δ_{pix} range observed across all three datasets.

Table 1: Ablation Study of CARLE Components (nuScenes Mini, CPU-only).

Variant	CR ↑	bpf (Kb) ↓	Lat. (ms)	Mem. (MB)
Frame-wise RLE	570.2	20.2	52.55	0.0024
Change-mask only (No Encoding)	1.0	11,520.0	60.40	1.3733
Change-mask + RLE	265.1	43.5	60.40	0.0052
Change-mask + CARLE (No Class Reduction)	130.0	88.6	60.40	0.0106
Change-mask + CARLE (Full)	**265.1**	**43.5**	**60.40**	**0.0052**

3 Experiments

3.1 Datasets and Evaluation Setup

Experiments are conducted under CPU-only edge constraints with no GPU acceleration. The nuScenes Mini dataset [15] consists of 2,424 frame pairs across 10 scenes and 6 cameras at 900×1600 resolution, using YOLO11n-seg as the upstream segmentation model ($B_{\text{raw}} = 11{,}520$ Kb/frame). The field-test dataset includes 1,073 frames from 4 real-world driving videos at 848×478 resolution (3,166.8 Kb/frame), covering static, mixed, and dense traffic conditions. The Cityscapes Panoptic Parts dataset [16], built on the Cityscapes benchmark [17], contains 1,487 consecutive ground-truth mask pairs across 18 cities at 1024×2048 resolution (16,384≈ Kb/frame). The implementation is publicly available at `https://github.com/Pritamresearch/carle-v2x`.

3.2 Ablation Study

Table 1 isolates the contribution of each pipeline component on nuScenes Mini.

Frame-wise RLE achieves 570.2× by exploiting within-frame spatial homogeneity, but retransmits all unchanged pixels on every frame. The change-mask-only variant confirms that the differential step alone is insufficient: without encoding, the sparse change map occupies the same size as the raw mask (CR = 1.0). Adding standard RLE recovers full compression (265.1×), demonstrating that temporal sparsity is the dominant factor. However, applying CARLE without the four-class reduction results in a CR of 130.0×. because the increased diversity of labels causes runs across labels runs across class boundaries. The complete pipelinepixel-wise detection, four-class conversion, and CARLEachieves 265.1× at the smallest memory (0.0052 MB).

3.3 Segmentation Backbone Sensitivity

Table **??** shows the unexpected result that lighter, lower-accuracy instance models have a *higher* CR than heavier, more-accurate semantic models. YOLO11n-seg (2.9M parameters, 29.7% mAP) achieves 265.1×, outperforming LRASPP-MobileNetV3 [18] (3.2M, 57.9% mIoU) at 196.2× and DeepLabV3-MobileNetV3 [19] at 191.0×. The mechanism is label map smoothness: lower-capacity models fewer class boundaries, producing bigger homogeneous regions, which lead longer runs

Table 2: Segmentation Backbone Comparison in the CARLE Pipeline (nuScenes Mini, 900×1600).

Model	Type	Params (M)	Size (MB)	Acc.*	CR ↑ (×)	bpf ↓ (Kb)	Lat. (ms)
YOLO11n-seg	Instance	**2.9**	**6.0**	29.7% mAP	**265.1**	**43.5**	60.4
YOLO11l-seg	Instance	27.6	55.7	43.2% mAP	225.5	51.1	112.7
LRASPP-MobileNetV3	Semantic	3.2	6.6	57.9% mIoU	196.2	58.7	**48.0**
DeepLabV3-MobileNetV3	Semantic	11.0	22.2	60.3% mIoU	191.0	60.3	104.4
YOLOv8n-seg	Instance	3.4	7.1	30.5% mAP	162.6	70.8	105.6
YOLOv8s-seg	Instance	11.8	23.9	38.9% mAP	146.8	78.5	243.1

Instance: mAP@50:95 on COCO val2017. Semantic: mIoU on Pascal VOC 2012. YOLO11n/YOLO11l/LRASPP/DeepLabV3: CoreML, Apple Neural Engine. YOLOv8: PyTorch CPU. Latency = inference + CARLE encoding.

Table 3: Compression Benchmark on nuScenes Mini (2,424 frames, 900×1600, CPU-only). ∼: values from original publications.

Method	CR ↑	bpf (Kb) ↓	Lat. (ms) ↓	Mem. (MB) ↓	mIoU ↑	Decoder
Raw Mask	1.0×	11520.0	0.00	1.37	1.00	No
Frame-wise RLE	458.8×	25.1	3.05	1.38	1.00	No
Delta Encoding	1.0×	11520.0	0.11	1.37	1.00	No
LZ4	167.1×	68.9	0.55	1.38	1.00	No
Snappy	21.0×	548.7	0.29	**0.07**	1.00	No
UCVC (2024) [20]	∼50×	∼230	∼116	∼480	∼0.91	Yes
DVC	∼27×	∼432	∼192	∼650	∼0.87	Yes
CARLE (Ours)	**265.1**×	**43.5**	60.40	2.75	**1.00**	**No**

of CARLE and fewer changed pixels K_t per differential step. This suggests a co-design guideline for edge V2X nodes: maximising segmentation smoothness rather than mIoU may may lead to more efficient communication.

4 Results and Discussion

4.1 Benchmark Compression Performance

Table 3 compares CARLE to spatial, lossless, and learned baselines on nuScenes Mini (same CPU-only settings) conditions. Latencies are end-to-end (inference marked ∼ are from original papers on other hardware.

There are three lessons. First, Delta Encoding achieves CR = 1.0 even when computing inter-frame differences: it works in the pixel domain, it yields a dense residual as compressible no more compressible than the original mask, proving that differencing alone, cannot rely on temporal differencing alone. Second, Frame-wise RLE achieves 458.8× via within-frame spatial homogeneity but retransmits all pixels every frame. CARLE transmits only changed-label entries, reducing bpf to 43.5 Kb/frame37% below LZ4 and 13× below Snapplyby exploiting label-space temporal sparsity. Third, learned codecs UCVC and DVC

Table 4: Per-Camera and Scene-Level Compression (nuScenes Mini, 900×1600, YOLO11n-seg, CPU-only).

Camera	**CR** (×)	**bpf** (Kb)	**Lat.** (ms)	**Scene**	**CR** (×)	**bpf** (Kb)	**Type**
CAM_FRONT	218.6	52.70	77.2	scene-1077	**1000**	**11.5**	Sparse road
CAM_FRONT_L	257.8	44.68	55.1	scene-0796	597	19.3	Low-traffic
CAM_FRONT_R	220.1	52.33	63.8	scene-0757	411	28.0	Suburban
CAM_BACK	300.5	38.34	62.2	scene-1094	342	33.7	Mixed
CAM_BACK_L	**414.1**	**27.82**	**46.0**	scene-0061	233	49.5	Urban
CAM_BACK_R	272.9	42.21	58.2	scene-0916	146	78.9	Dense int.

achieve lower CR (50× and 27×), require a decoder, consume two to three orders of magnitude more memory, and incur higher latency (∼116192 ms vs 60.40 ms), while sacrificing fidelity (mIoU ∼0.870.91 vs 1.00). CARLE is the only method achieving lossless fidelity, no decoder, and sub-100 ms latency simultaneously on a CPU-only node.

4.2 Per-Camera and Scene-Level Analysis

Table **??** presents the performance of CARLE viewpoint and scene type in the 10 nuScenes Mini scenes. The back cameras perform better than the front cameras CAM_BACK_LEFT reaches 414.1× vs 218.6× for CAM_FRONTbecause the forward camera view has the most of oncoming objects and lane change transitions, resulting in more changed pixels per differential step. This directional CR asymmetry enables per-stream prioritization for multi-camera V2X pipelines. The 6-camera bandwidth of 258 Kb/frame is a 99.6% compression of the 69,120 Kb/frame raw data, which is compliant with 3GPP NR-V2X sidelink. Scene-level CR spans 146× to 1000× across the 10 scenes with no rate-control tuning, and that bandwidth grows with semantic activity.

4.3 Real-World Field-Test Results

Table 5 gives CARLE results on 1,073 frames from 4 videos of resolution 848×478. Video 4 (170×, 38.58 Kb/frame, $\delta_{\text{pix}} = 0.955\%$) is the closest closest to nuScenes conditions and gives similar performance: CARLE has the same bpf (38.58 vs 43.5 Kb/frame) with a similar semantic load. The mean latency of 17.97 ms is 3.4× faster than nuScenes, due to the smaller due to smaller frame sizes, not different algorithms, and 99.5% of frame encode in 30 mswell below the 3GPP deadline of 100 ms.

4.4 Cross-Dataset Validation and Cityscapes Analysis

Table **??** and Fig. 4 consolidate CARLE performance across all three datasets. On Cityscapes (1024×2048, 1,487 pairs), adjacent ground-truth pairs yield 179,346×

Table 5: Field-Test Results (848×478, raw = 3,166.8 Kb/frame, CPU-only).

Video	**CR ↑**	**bpf (Kb) ↓**	**Lat. (ms)**	δ_{pix} **(%)**
Video 1 (Static road)	67557×	0.05	14.90	0.000
Video 2 (Mixed traffic)	9731×	4.06	17.53	0.244
Video 3 (Mixed + sparse)	35913×	5.97	16.19	0.218
Video 4 (Dense intersection)	170×	38.58	23.40	0.955
Overall (all diff)	**33441×**	**12.70**	**17.97**	0.350
Active frames only	**884×**	**24.78**	**20.91**	

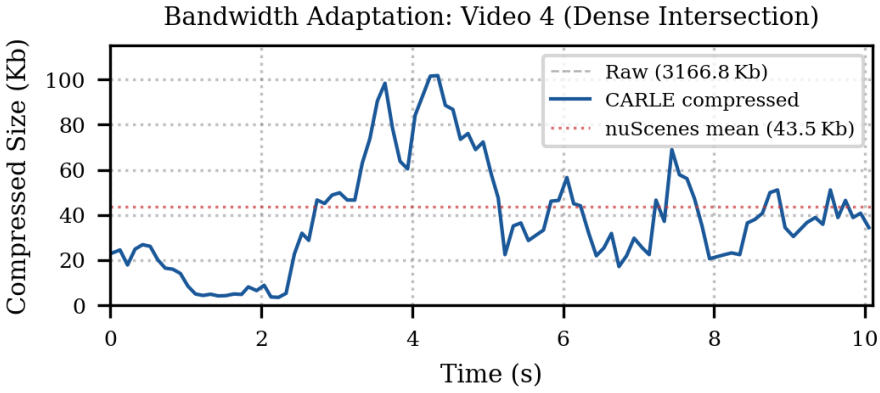


Fig. 3: Real-time bandwidth on field-test Video 4. Four phases: start of traffic (01 s), inactivity (1.12.3 s), peak activity at 101 Kb/frame (2.44.5 s), and recovery (510 s). No rate-control signal used.

Table 6: Cross-Dataset Summary of CARLE Performance.

Dataset	**Res.**	**Pairs**	**CR (×)**	**bpf (Kb)**	**Lat. (ms)**	δ_{pix} **(%)**
Cityscapes GT	1024×2048	1,487	**179,346**	**0.093**	179.44	0.007
Field test (active)	848×478	547	884	24.78	20.91	0.350
nuScenes Mini	900×1600	2,364	265	43.01	60.40	4.500

CR at 0.093 Kb/frame, consistent with their near-zero δ_{pix} (0.007%). Low CR variability across 18 cities (std <3% of mean) suggests performance is governed by temporal change rate rather than scene content or geographic region. The mean latency of 179.44 ms exceeds the 3GPP constraint, attributable to the 1.44× larger pixel count vs nuScenes; the tight latency distribution (std 25.52 ms vs 51.88 ms) confirms encoding cost is near-constant when $\delta_{\text{pix}} \to 0$.

Fig. 4 compares the three datasets to the theoretical curve CR $\propto 1/\delta_{\text{pix}}$ from Eq. (5) (not fitted). All three points are close to this curve over four orders of magnitude in CR and three in δ_{pix}, covering various resolutions, capture methods

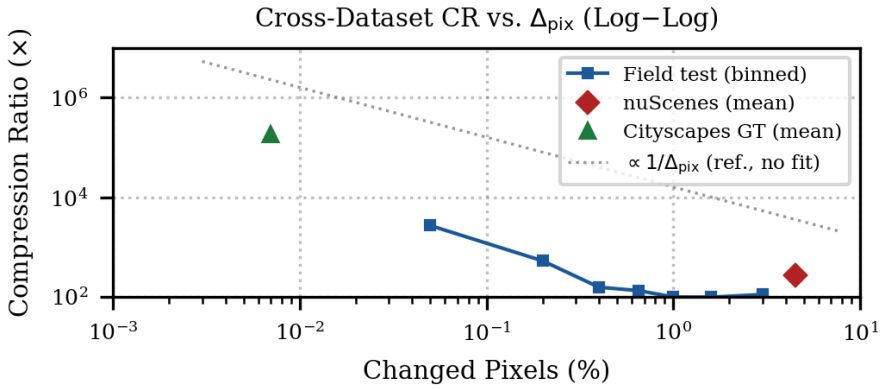


Fig. 4: Cross-dataset CR vs. δ_{pix} (loglog). All three operating points lie close to the reference curve (no fitting), spanning four orders of magnitude in CR across datasets differing in resolution, modality, and region, supporting the theoretical model of Eq. (5).

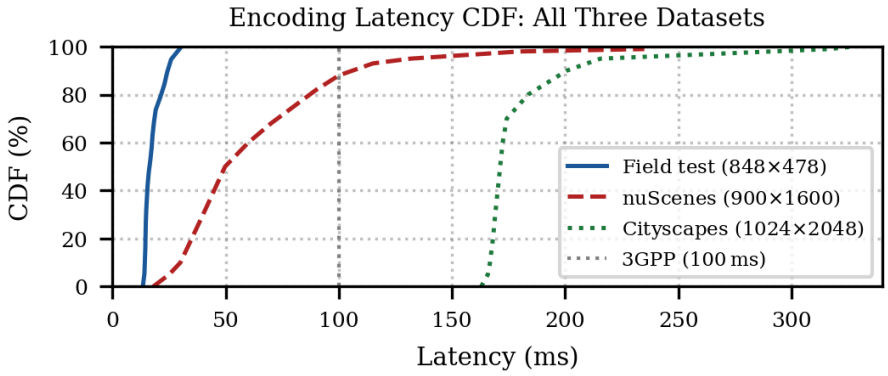


Fig. 5: Latency CDF across all three datasets. Latency follows resolution order regardless of CR. Field test and nuScenes satisfy the 3GPP 100 ms constraint for >99.9% and >94.3% of frames; Cityscapes requires hardware acceleration at 1024×2048.

and locations. While three datasets do not establish universality, the consistency across such diverse conditions supports the theoretical model and indicates that CR is primarily determined by δ_{pix} rather than dataset-specific factors.

4.5 Encoding Latency Analysis

Encoding latency scales with image resolution rather than CR: field test (848×478, mean 17.97 ms), nuScenes (900×1600, 60.40 ms), and Cityscapes (1024×2048, 179.44 ms) follow a strict resolution hierarchy despite spanning four orders of magnitude in CR, confirming that CARLE's cost is dominated by the $\mathcal{O}(HW)$ pixel-wise comparison and run-length scanning steps rather than encoded run count.

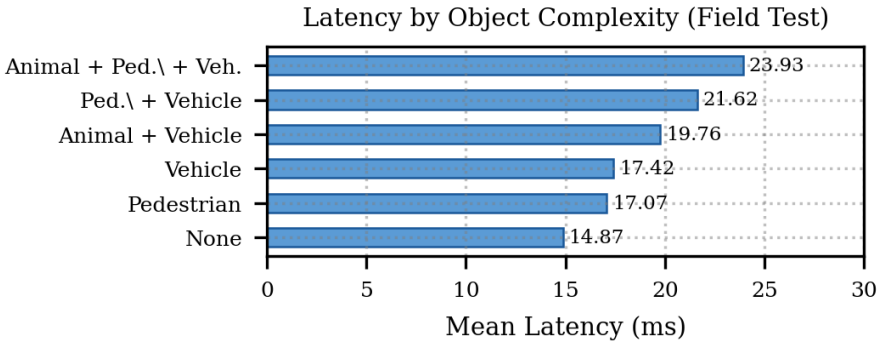


Fig. 6: Mean latency by object class combination (field test, 1,069 frames). Latency scales monotonically with semantic complexity (14.8723.93 ms), remaining within the 3GPP 100 ms budget.

Field test and nuScenes satisfy the 3GPP 100 ms constraint for 99.9% and 94.3% of frames respectively (Fig. 5); Cityscapes fails entirely at CPU-only throughput, establishing a resolution ceiling and motivating hardware-accelerated encoding as future work. Field-test latency increases monotonically with semantic complexity (Fig. 6): from 14.87 ms (no objects) to 23.93 ms (vehicle + pedestrian + animal)an 8.06 ms range entirely within budgetreflecting the positive correlation between object count and K_t.

5 Conclusion

This paper presented CARLE (Class-Aware Run-Length Encoding), a deterministic, training-free framework that compresses semantic segmentation masks for V2X collaborative perception by transmitting only inter-frame label changes as startlengthclass triplets, with unchanged pixels set to a reserved null symbol (0), enabling direct receiver-side map updates without decoding. Evaluated across three CPU-only datasets2,424 nuScenes Mini frames, 1,073 field-test frames, and 1,487 Cityscapes panoptic pairsCARLE achieves 265.1× at 60.40 ms, 884×

at 17.97 ms, and 179,346× at 179.44 ms, all with mIoU = 1.00 and no parameter tuning. A combined 6-camera stream requires only 258 Kb/frame (99.6% reduction), meeting 3GPP NR-V2X sidelink constraints. Future work targets hardware-accelerated encoding, multi-vehicle scenarios, and compressed-domain inference on the CARLE triplet stream.

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